Myungsin Cho

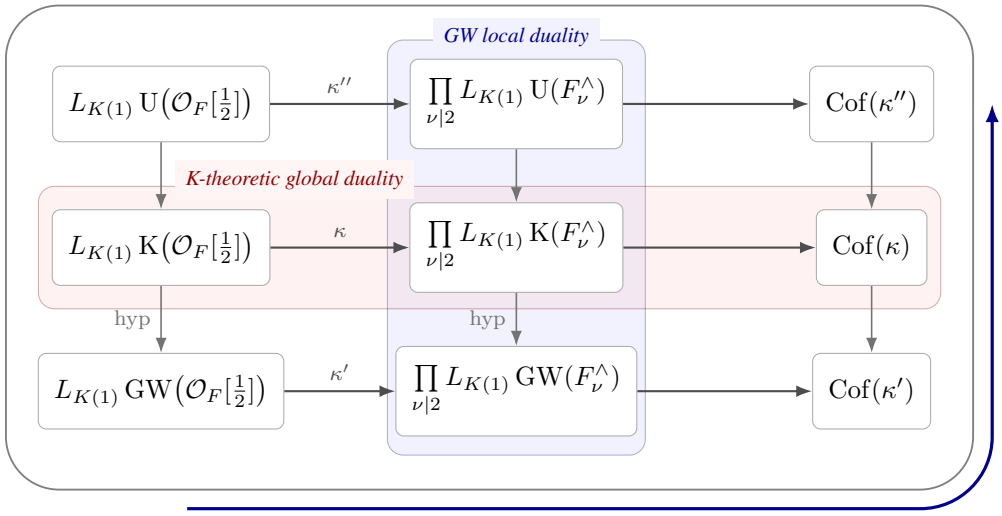
Research Statement

My research is in stable homotopy theory, with connections to algebra and number theory. One direction studies duality in algebraic K-theory, with the long-term goal of understanding duality and descent in Real algebraic K-theory. A second direction concerns equivariant stable homotopy theory and equivariant algebra, where incorporating symmetry leads to new algebraic structures. These interests also provide starting points for undergraduate research.

Duality in algebraic K-theory

In linear algebra, a linear map can be studied through its action on vectors or through its action on linear functions. Passing to the dual reverses the direction of the map while preserving the information it carries. My research seeks similar phenomena in algebraic K-theory. This theory studies linear algebra over rings and packages information about modules into topological invariants. Although these invariants are difficult to compute, duality can reveal how apparently different pieces of information determine one another.

My dissertation studies such a relationship in arithmetic. Comparing global and local K-theory gives three related objects: $L_{K(1)}\,\mathrm{K}\big(\mathcal{O}_F[\tfrac{1}{2}]\big)$ the global theory , $\prod_{\nu|2} L_{K(1)}\,\mathrm{K}(F_\nu^\wedge)$ the local theory, and $\mathrm{Cof}(\kappa)$ the information measuring their difference. Duality reverses this comparison. The global theory and the comparison term exchange roles, while the local theory is self-dual. I establish this symmetry at the prime 2, completing the K-theoretic Tate–Poitou duality previously proved at odd primes by Blumberg and Mandell. The red row of the diagram below displays this relationship.



Anderson duality: 180° rotation

Linear algebra provides another familiar starting point for extending this work. A vector space may carry a bilinear form, such as an inner product, which adds structure beyond the vector space itself. Grothendieck–Witt theory records this form structure, while Real algebraic K-theory brings it together with ordinary K-theory through symmetry. With Lucy Yang, I establish a local duality relating ordinary K-theory, GW-theory, and their comparison theory U. This gives the symmetry shown in the blue column. Our ongoing global work asks whether these two reversals fit together into a half-turn symmetry of the full diagram.

My long-term goal is to understand these dualities as parts of a single relationship in Real algebraic K-theory. I also seek an explanation of why this relationship exists. In ordinary K-theory, descent connects the theory to arithmetic cohomology, where classical duality is well understood. A corresponding explanation

incorporating symmetry is not yet available. I use the known and expected dualities to guide the search for such a Real descent theory.

The same duality has an application to the K-theory of the sphere spectrum, a basic building block of stable homotopy theory. The cyclotomic trace relates this difficult invariant to a more accessible one. My theorem shows that the information missed by this approximation is itself described by a duality with the arithmetic K-theory of the integers. Thus arithmetic duality also explains a relationship between two invariants arising in topology.

The sphere-spectrum application also shows how trace methods enter this program. The trace relates two invariants, and duality describes the information between them. This connection motivates my study of Real trace invariants, which incorporate symmetry into these relationships. With Gerhardt, Keenan, Moreno, and Quigley, I develop Real homological trace methods that account for this symmetry and its additional algebraic operations. We obtain explicit results for Real bordism theories, extending the methods of Bruner and Rognes to the Real setting. A natural next step is a Real Segal conjecture for these theories, toward understanding Real trace invariants and ultimately Real algebraic K-theory beyond arithmetic examples.

Equivariant stable homotopy theory and algebra

Symmetry also changes familiar constructions in algebra. For example, in a commutative ring, if a symmetry exchanges two elements, their sum and product are unchanged. These elementary constructions suggest additive and multiplicative ways of combining information while respecting symmetry. In equivariant algebra, they develop into operations called transfers and norms. There can be many such operations, corresponding to different kinds of symmetry. My research asks which of them can be used together and how they must interact.

With Chan, Mehrle, Sanchez Ocal, Osorno, Szczesny, and Verdugo, I study this interaction through combinatorics. As with ordinary addition and multiplication, the operations must obey compatibility rules resembling a distributive law. We encode the allowed operations by transfer systems and ask whether compatible choices can be realized as coherent systems of operations, organized by operads. Our results realize several classes of compatible choices, and we conjecture that compatibility is sufficient in general. The goal is to understand when a combinatorial description of the desired operations can be turned into a mathematical structure carrying them.

With Noah Wisdom, I study how symmetry changes the gradings that organize topological invariants. For order-two symmetry, we show that some apparently additional grading information is already determined by the underlying algebraic structure.

Undergraduate research I want students to learn to formulate their own mathematical questions. Familiar concepts offer a starting point. What changes if we alter a condition in a definition? What happens if we bring together two concepts usually studied separately? Equivariant rings offer an example, combining rings and symmetry to explore natural definitions, examples, and properties. The structures and invariants of manifolds with symmetry offer another direction. I would guide students in turning their questions into problems they can investigate through concrete examples. As patterns emerge, we would refine the questions, formulate conjectures, and seek proofs or counterexamples. My aim is for students to experience how a question develops through investigation and to take increasing responsibility for the direction of their work.